Sang Hak Lee

Marine Data Engineer · Vessel & weather data collection, datasets and quality

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Data engineer who **collects and cleans** AIS / V-PASS vessel data and marine-weather forecast data (NetCDF), and **builds datasets** on PostgreSQL data models I design. Embeds spec-based **data quality validation** and processing **metadata management** into pipelines; licensed weather forecaster with meteorology and ocean-environment certifications, bringing domain understanding of vessel, weather and route data.

EXPERIENCE

GeoSR (Geo System Research) · Data Engineer

Feb 2023 - Present

- **Vessel data collection & cleansing** — Cleaned TB-scale AIS trajectories for the Busan area (outliers, duplicates, time alignment) and joined route/anchorage data to build anchoring, loitering and route-deviation candidate datasets (GeoPandas, multiprocessing).
- **V-PASS dataset construction** — Streaming merge and filtering of vessel observations (data years 2023-2025) into regional/grid statistical datasets and density maps, joined with catch-grade data.
- **Weather & ocean data pipeline** — Automated NetCDF/CSV ingestion → spec validation → raster and current-vector conversion → PostgreSQL loading → GeoServer publishing (Python, Shell, Docker, air-gapped).
- **Data modeling** — Split loading for 11 monitoring/forecast products from 4 institutions into a **file catalog** (data type, year-month, monitoring/forecast, publication flag), **validation results** (per-file pass/fail and error items) and **value tables** (current lat/lon, speed, direction, depth), linked by file name.
- **Data quality validation** — Automated spec checks (Class, Variable, Level, Unit) with error records, exclusion and count reconciliation, service-point to model-grid matching, old vs. new product comparisons.
- **Metadata management** — Maintained processing/error history, input spec sheets and data specifications; traced source file → validation result → loaded values per file; SHA-256 change detection re-validates only changed files.
- **Systems & documentation** — Built forecast production servers (PostgreSQL), scheduling, monitoring, incident response; requirements and test documents; conference poster (2024), workshop talk (2025).

Korea Meteorological Industry Promotion Agency · R&D Planning Intern

Jun - Sep 2022

- Researched weather technology and policy trends; wrote monthly newsletters and quarterly reports.

SELECTED PROJECTS

Ocean Forecast GeoPipeline

2025.06 - 2026.04

NetCDF/CSV spec validation → spatial conversion → PostgreSQL loading → GeoServer publishing.

AIS Trajectory Analytics

2024 - 2025

Activity-candidate detection from AIS tracks with route and anchorage polygons; KOSMEE autumn conference poster (2024) and workshop talk (2025).

Marine Travel AI Agent · GeoSR internal R&D

2026

Managed public data consumed by an LLM agent — automated TourAPI and 8 marine-forecast feed collection (n8n), duplicate-ID, coordinate-range, missing-page and per-feed freshness checks, collection/issue timestamps, typed validation tools for LLM use (LangGraph, FastAPI).

SKILLS

Data	Python, SQL, PostgreSQL, Pandas, GeoPandas
Formats	NetCDF, AIS, V-PASS, GeoTIFF, GeoServer, QGIS
Infra	Linux, Shell, Docker, scheduling & monitoring
Other	Fortran, Rust, LangGraph, FastAPI, Ollama (LLM agents)

CREDENTIALS · EDUCATION

Weather Forecaster License · Engineer Meteorology · Engineer Ocean Environment

B.S. Earth & Environmental Sciences, Jeonbuk National University (2016 - 2022)

Research Club Grand Prize, JBNU College of Natural Sciences (2021)

① Monitoring & forecast product loading pipeline

GeoSR · Jun 2025 - Apr 2026

Air-gapped pipeline that validates 11 monthly marine/weather monitoring and forecast products (NetCDF, CSV) from 4 institutions (KIOST, NIFS, KHOA, KOPRI), loads them into PostgreSQL and publishes GeoServer layers.

Role Data model design, spec validation, spatial conversion and ETL; specifications and test documents

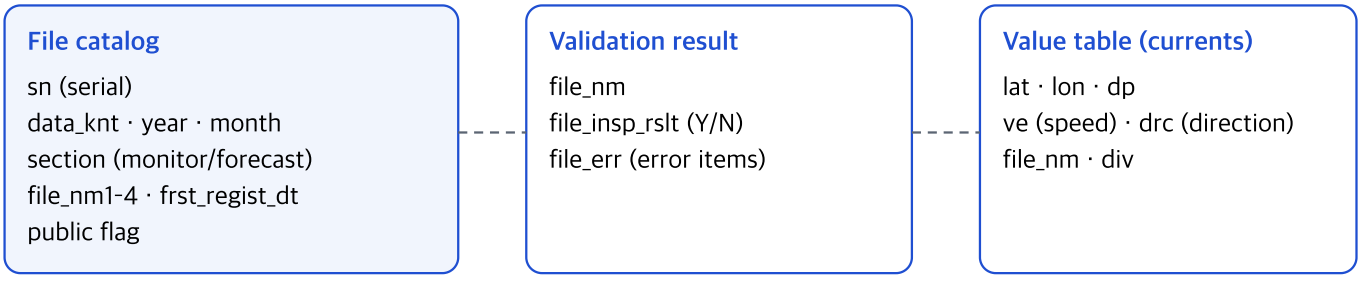
Design Separated product **metadata** (data type, year-month, monitoring/forecast, registration time, publication flag), **validation results** and **values** into distinct tables; the catalog is queried and updated by (data type, year, month, section), while results and values link by file name to trace source → validation → load

Validation Automatically compared Class, Variable, Level (depth) and Unit against the input spec; recorded pass/fail and error items per file

Processing File watching (Python, Shell, Docker), raster regridding/reprojection, speed and direction from U/V, upserts, transactions, reprocessing, scheduling

Why A separate validation table makes per-file failure causes queryable; file-signature comparison reduces repeated spec validation of unchanged files

Results Item-level reports, GeoTIFF and current-vector CSV, GeoServer layers



Loading structure (summary of actual tables)

Change detection & incremental runs Kept a processing log of size, mtime and SHA-256 per file (filename, size, mtime, sha256, processed_at) and re-ran spec validation only for changed files.

② AIS vessel data cleansing & activity detection

GeoSR · 2024 - 2025

Combined historical Busan-area AIS with vessel metadata and route/anchorage polygons to detect and visualize anchoring, loitering and route-deviation candidates.

Role Trajectory preprocessing, spatial-rule and time-alignment detection, parallel visualization; poster and workshop talk

Cleansing Removed outliers and duplicates from TB-scale raw data, aligned tracks per vessel in time, derived speed, heading and displacement

Design Judged candidates with **readable rules** combining polygon membership and movement features so each decision can be reviewed

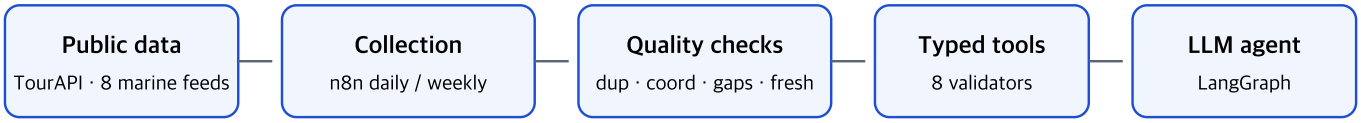
Processing GeoPandas spatial joins; multiprocessing for large map time-lapse generation

Results KOSMEE 2024 autumn conference poster, Jan 2025 workshop talk

③ Marine Travel AI Agent — public data for an LLM

GeoSR internal R&D · 2026

Collects and validates tourism, marine-forecast, weather and transport public data so an LLM agent can cite checkable evidence. Quality management of **external data an LLM consumes**, not training-dataset construction.



Role	Public-data integration, collection workflows, quality checks and typed validation tools, itinerary rule engine
Quality	Content-ID duplicates, coordinate range, missing/repeated pages, per-feed freshness; atomic refresh keeps prior data on failure
Metadata	Stored collection time, forecast issue time and source with each record so the time basis of an LLM answer can be checked
LLM link	In a LangGraph generate → validate → user choice → recheck flow, the LLM reads data through typed validation tools; SQLite checkpoints restore interrupted plans
Scale & tests	~10K tourism records, 8 marine forecast feeds; 195 Python and 28 Node tests passing; Langfuse execution tracing

④ V-PASS vessel observation datasets

GeoSR · data years 2023 - 2025

Role	Built the full batch workflow from vessel-observation selection, aggregation and spatial joins to consistency checks
Processing	Annual streaming merge, speed and vessel-ID filters, monthly (36) and complete-season (11) aggregation, regional grid assignment and count sums, weighted Quartic KDE, catch-grade join
Join rules	Exact match of CSV grid numbers to grid-point IDs; duplicate CSV keys summed, duplicate spatial IDs rejected; vessel IDs compared verbatim without decoding
Consistency	Count reconciliation by **input = assigned + unmatched + outside**; with overlapping regions, regional sums split into unique counts and overlap duplication
Quality	Malformed records quarantined with offsets and reasons; mean denominators exclude missing/NULL but include zero; methodology and validation records public

⑤ Forecast system build, operation & analysis

GeoSR · 2023 - 2025

Build	Forecast-index and sea-condition chart production servers: quotes, install, remote access, packet filtering, PostgreSQL, Python/Java; backup NAS (2025)
Operation	Production scheduling, daily output monitoring, incident response and reporting
Analysis	Old vs. new model product comparisons, severe-weather-day and website/OpenAPI usage analysis, monthly statistics and summaries